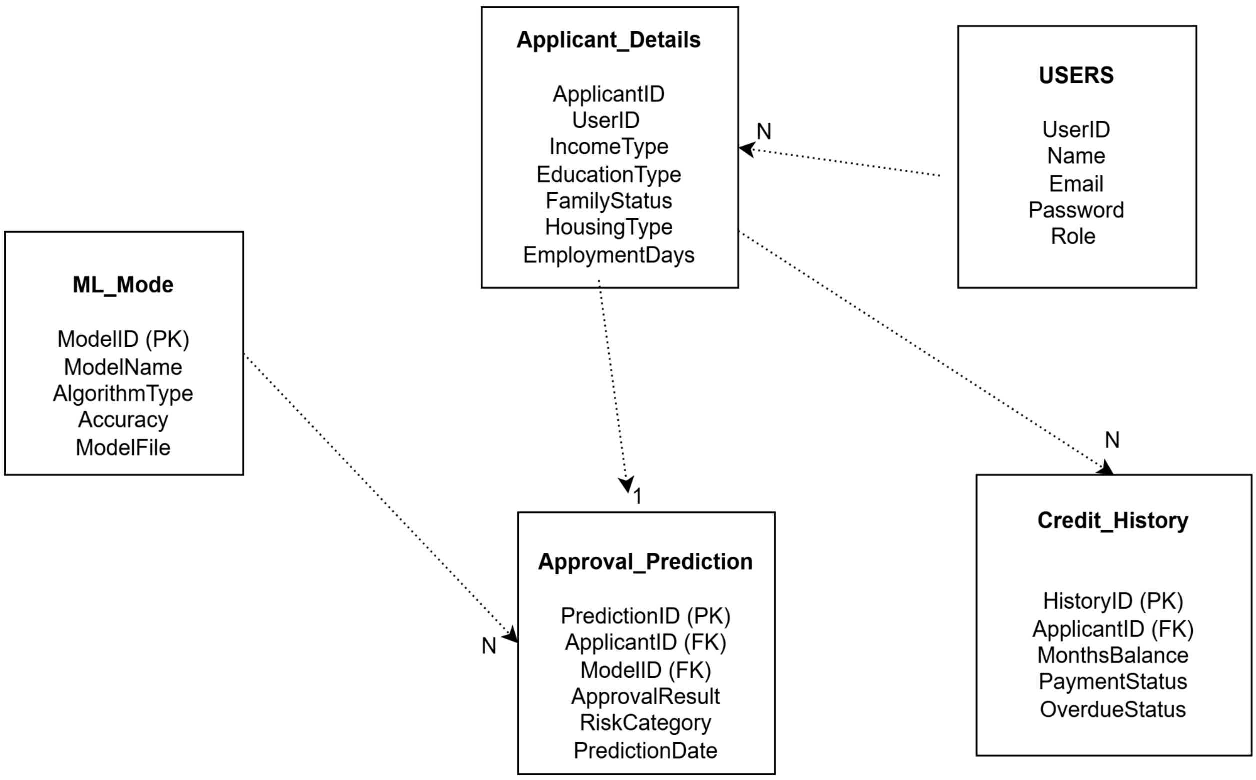
Entity Relationship Diagram for Credit Card Approval Prediction System

Introduction

The Credit Card Approval Prediction System is a machine learning-based application designed to automate the process of evaluating credit card applications. The system analyzes applicant information such as income type, education level, employment status, housing details, and credit history to predict whether a credit card application is likely to be approved or rejected.

The Entity Relationship Diagram (ERD) provides a visual representation of the database structure used in the system. It illustrates the entities, attributes, primary keys, and relationships between different components such as users, applicant details, credit history records, machine learning models, and approval predictions. The ERD helps in organizing data efficiently and ensures proper interaction between various modules of the application.

This database design serves as the foundation for storing applicant information, processing credit history data, generating predictions using machine learning algorithms, and managing prediction results within the Credit Card Approval Prediction System.



Entities

1. Users

The Users entity stores information about users who access the system.

Attributes:

- UserID (Primary Key)
- Name
- Email
- Password
- Role

2. Applicant_Details

The Applicant_Details entity stores personal and financial information of applicants.

Attributes:

- ApplicantID (Primary Key)
- UserID (Foreign Key)
- IncomeType
- EducationType
- FamilyStatus
- HousingType
- EmploymentDays

3. Credit_History

The Credit_History entity stores applicant credit records and payment history.

Attributes:

- HistoryID (Primary Key)
- ApplicantID (Foreign Key)
- MonthsBalance
- PaymentStatus
- OverdueStatus

4. ML_Model

The ML_Model entity stores machine learning model details used for prediction.

Attributes:

- ModelID (Primary Key)
- ModelName
- AlgorithmType
- Accuracy

- ModelFile

5. Approval_Prediction

The Approval_Prediction entity stores the prediction results generated by the machine learning model.

Attributes:

- PredictionID (Primary Key)
- ApplicantID (Foreign Key)
- ModelID (Foreign Key)
- ApprovalResult
- RiskCategory
- PredictionDate

Relationships

Users → Applicant_Details

One user can manage multiple applicant records.

Relationship**:** One-to-Many (1:N)

Applicant_Details → Credit_History

One applicant can have multiple credit history records.

Relationship**:** One-to-Many (1:N)

Applicant_Details → Approval_Prediction

One applicant receives one final approval prediction.

Relationship**:** One-to-One (1:1)

ML_Model → Approval_Prediction

One machine learning model can generate predictions for multiple applicants.

Relationship**:** One-to-Many (1:N)

Cardinality

- One User → Many Applicants
- One Applicant → Many Credit History Records
- One Applicant → One Approval Prediction
- One ML Model → Many Approval Predictions